
Distributed genetic algorithm for lifetime coverage optimisation in wireless sensor networks

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Abstract: In this paper, a protocol called distributed genetic algorithm for lifetime coverage optimisation (DiGALCO) is suggested to preserve the coverage and enhance the lifetime of a wireless sensor network (WSN). DiGALCO combines three energy-efficient schemes: virtual network subdivision into subfields, distributed cluster head selection in each subfield, followed by sensor activity scheduling-based genetic algorithm (GA) optimisation performed by each cluster head. DiGALCO works into rounds. Each round consists of three phases: discovery, cluster head selection and GA decision and sensing. The decision process, which results in an activity scheduling vector, is achieved by a cluster head node through executing the GA to pick out a set of sensors staying active for monitoring through the current sensing round. In comparison with other protocols, several experimental results were done by using OMNeT++ network simulator show that DiGALCO is capable of prolonging the lifetime of WSN and gives enhanced coverage performance.

Keywords: wireless sensor networks; WSN; coverage; network lifetime; genetic algorithm; scheduling.

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1 Introduction

The advances in the micro-electronics and wireless communications technologies have led to emergence of a new type of networks: wireless sensor networks (WSNs). This type of networks consists of a large tiny low-cost devices (called sensors) deployed in the area of interest to be monitored. A sensor node can sense, process and forward data over a wireless communication to a sink node. WSNs can be used in several applications like smart agriculture, healthcare, nature environment protection, transportation, smart cities, military, etc. (Akyildiz et al., 2002; Yick et al., 2008). WSN represents as one of the most important technologies for the future (Rawat et al., 2014). The coverage problem is one of the essential research topics in WSNs (Fan and Jin, 2010). In WSNs, the quality of service (QoS) of the sensing task is measured by coverage. The coverage refers to how well the area of interest is monitored by the deployed nodes with a minimum amount of energy (Zhu et al., 2012). In the published literature, three types of coverage can be found (Ghosh and Das, 2008). Area coverage is to cover all the area of interest. Target coverage is to cover a limited number of points or targets in the area of interest. The main objective of barrier coverage is to detect an event when it try to cross a barrier such as intrusion detection or border surveillance. This article focuses on the area coverage.

It is beneficial to deploy a large number of sensors in the sensing field because of the limited life of sensor node and the difficulty of replacing the batteries in remote environments. WSN can exploit the redundant sensors to extend the network lifetime. In dense WSN, it is unnecessary to turn on all sensors at the same time. The WSN lifetime would be reduced if all the sensor nodes are turned on at a time. Scheduling scheme is required to put some sensors in an active state to provide sensing services and put other sensors in a sleep state to save power thus increase the lifetime of WSN. The main advantages of scheduling the sensors are to activate a less number of sensors, increase the battery life of the sensor node, decrease the collision on the channel and extend the WSN lifetime (Yu et al., 2012).

This paper makes the following contributions:

- 1 The work presented in this article concentrates on the area coverage problem, with the aim of improving the lifetime of WSN by using an energy-efficient scheduling-based genetic algorithm (GA) optimisation. The proposed protocol is called distributed genetic algorithm for lifetime coverage optimisation (DiGALCO), conserves the coverage and extends the lifetime of WSNs. DiGALCO protocol relies on the framework of the distributed lifetime coverage optimisation (DiLCO) protocol (Idrees et al., 2015). The main idea of this framework is to exploit spatial and temporal subdivision. The sensing field is virtually partitioned into smaller subfields.

After that, the lifetime is subdivided into rounds inside each subfield. This way of grouping of sensor nodes is similar to clustering. Inside each subregion, the sensor nodes cooperate with each other to select the cluster head which will apply the GA to provide the best representative set of sensor nodes to take the responsibility of sensing during the current round. DiGALCO protocol uses a distributed efficient way to select the cluster head inside the subfield.

- 2 DiGALCO protocol uses a GA instead of optimisation solver in order to schedule nodes' activities periodically inside the subregion. The coverage problem formulation is modified to consider nodes centres instead of the primary points during solving the optimisation problem.
- 3 The discrete event simulator OMNeT++ is used to conduct the experimental results to illustrate the efficiency of DiGALCO protocol. The DiGALCO protocol has been compared to four methods found in the related works: DESK (Vu et al., 2006), GAF (Xu et al., 2001), DiLCO protocol published in Idrees et al. (2015) and also to the PeCO protocol (Idrees et al., 2016). DiLCO and PeCO use the same framework as DiGALCO but they are relayed on GLPK optimisation solver for sensor scheduling.

The rest of this article is organised as follows. Section shows the related literature in the field. Section 3 is allocated to DiGALCO protocol illustration. Section 4 exhibits the experimental results conducted using OMNeT++ simulator (Varga, 2003). Finally, we provide conclusion and future works in Section 5.

2 Related works

Recently, the coverage problem has been received a high attention, which concentrates on how the physical space could be well monitored after the deployment. Coverage is one of the QoS parameters in WSNs, which is highly concerned with power depletion Zhu et al. (2012). The related works of the coverage problem are summarised in this section and the shortcomings of reviewed approaches are being summarised.

Xu et al. (2001) proposed a geography-informed energy saving routing algorithm for ad hoc network, called GAF. In this algorithm, nodes use geographic information of location to partition the sensing field into fixed small cells. Each cell remains with a constant size. Every sensor inside a cell has two states either sleep or listen. There is only one sensor node stays awake within each fixed cell in order to perform the data sensing and communication tasks. The GPS is used by each sensor node to join itself with a certain cell. Vu et al. (2006) introduced a distributed energy-efficient scheduling algorithm for K-coverage in WSNs, called DESK. DESK operates into rounds. The algorithm requires only onehop neighbour information. With DESK, every sensor determines its state (Sleep or Active) depending on the perimeter coverage model that presented by Huang et al. (2005).

Li et al. (2011) presented a framework to convert any complete coverage problem to an incomplete coverage with a certain ratio of coverage by means of executing an algorithm to provide sets with a complete coverage and converting the sets of full coverage to sets of partial coverage via modifying the range of sensing. Özdemir et al. (2013) employ a recently developed multiobjective optimisation algorithm, named

multi-objective evolutionary algorithm based on decomposition (MOEA/D) to maintain the coverage and save the energy simultaneously in WSNs. The aim of the work presented by Rebai et al. (2014) is to provide a full coverage for a grid by spreading the required sensor nodes whilst preserving connectivity between the sensor nodes and a sink. A branch and bound (B&B) method is proposed to optimally resolve this problem. Mostafaei and Shojafar (2015) presented a new and effective approach for cover set generation depending on imperialist competitive algorithm (ICA). In order to observe all the targets in the sensing field, the sensors that have to be chosen in several cover sets are determined by ICA.

A learning automata approaches are introduced in the works (Mostafaei and Meybodi, 2013; Mohamadi et al., 2013; Mostafaei et al., 2015) to treat the problem of coverage in sensor networks. In order to extend the lifetime of the sensor network, the nodes are scheduled in several sets to cover the sensing field. Amgoth and Jana (2015) suggested an algorithm named distributed energy and coverage-aware routing (DECAR) to monitor the sensing field for the longest time as possible in the WSNs. Effective clustering and cluster head election approaches are used by the algorithm for routing. Bara'a et al. (2015), the authors introduce a multiobjective disjoint reliable sensor covers problem (DRSC). Two multiobjective metaheuristic approaches (called MOEA/D and NSGAI) are used to deal with this problem. They suggested using one heuristic approach instead of the crossover and mutation operators. The heuristic method incorporates in its mechanism a problem-specific knowledge. Results show that the positive effect of suggested heuristic method on the total performance of applied MOEAs. Abdulhalim and Bara'a (2015) presented a multi-layer metaheuristic approach to maximise the set covers number and minimise the overall number of nodes utilised by every set cover. The reliability of coverage of every cover set is improved by using postheuristic operator that receive the result of the multi-layer GA.

Idrees et al. (2015) suggested a new protocol named DiLCO to save the energy and keeps the coverage in WSNs. DiLCO represents an improvement on the previous work introduced by Idrees et al. (2014). DiLCO protocol is based on two stages. The sensing field is virtually divided into smaller subregions in the first stage whilst the DiLCO protocol is distributed inside every node in the network in the second stage. The proposed protocol performs an election method to select the leader in each small subregion. After that, the sensor node leader executes an optimisation solver for resolve the node activity scheduling optimisation to select the best group of nodes that provide a good coverage and improve the WSN lifetime. DiLCO is periodic and it includes four steps: exchange of information, selection of the leader, sensor schedule production decision and monitoring. The simulation results illustrate that DiLCO provides better performance compared with other existing methods. Idrees et al. (2016) presented a new protocol named PeCO that based on a new proposed coverage optimisation model. PeCO protocol uses an integer program to provide an optimal sensor schedule based on the perimeter coverage model and instead of using points as in DiLCO. The results show that PeCO can improve the network lifetime and maintain the coverage for a longest time compared with other methods.

Several algorithms to maintain the coverage and maximise the network lifetime were proposed in Cardei and Wu (2006), Wang (2011), Zhu et al. (2012), Valera et al. (2014), Liang et al. (2014) and Sangwan and Singh (2015).

2.1 Shortcomings

In spite of many energy efficient protocols for maintaining the coverage and improving the network lifetime in WSNs were proposed, non of them ensure the coverage for the sensing field with optimal minimum number of active sensor nodes and for a long time. For example, in a full centralised algorithm, optimal solutions can be given by using optimisation approaches, but in the same time, a high energy is consumed due to algorithm execution time and the communications among the sensors in the sensing field. Therefore, the full centralised approaches are not good candidate to use them especially in large WSNs. Whilst, a full distributed algorithms can not give optimal solutions because they use only local information of the neighbouring sensors, but in the same time, the energy consumption during the communications and executing the algorithm is highly lower. Whatever the case, this would result in a shorter lifetime coverage in WSNs.

2.2 Our protocol

In this article, DiGALCO in WSNs, called DiGALCO is suggested. The sensing field is partitioned into small subregions and a DiGALCO protocol is distributed in each sensor in the subregion. The WSN lifetime in each subregion is divided into rounds, each round includes three stages: Discovery, Cluster election, sensor node activity scheduling-based GA optimisation and sensing. The cluster heads are elected in an independent, asynchronous and distributed way in all the subregions of the WSN. After that, energy-efficient activity scheduling mechanism-based GA optimisation is performed by each cluster head in the subregions. GA optimisation is based on the sensor nodes centres in order to produce the optimal cover set of active sensor nodes, which are responsible of sensing during the current round. DiGALCO protocol merges and uses the main advantages of the centralised and distributed approaches and avoids the most of their disadvantages.

3 DiGALCO protocol description

In this section, DiGALCO protocol is presented that distributed in each sensor node in the network. It is based on three energy-efficient techniques: virtual network subdivision, cluster head election and sensor activity scheduling-based GA optimisation for ensuring sensing coverage while minimising energy consumption, applied periodically to efficiently improve the network lifetime.

3.1 Network model

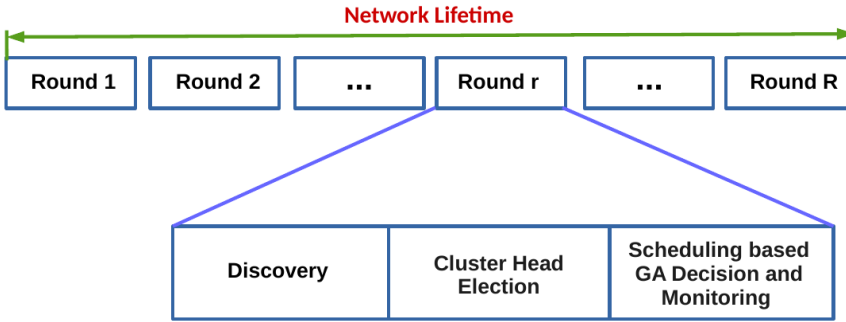
The homogeneous static sensor nodes are deployed in a uniform and random way in the area of interest. The communication range (R_c) and sensing range (R_s) are the same for all the sensor nodes. Each sensor node started with a different level of energy. The sensor network is dense to guarantee a full coverage for the sensing field. Every sensor node determines its position based on a software (localisation algorithm) or hardware (GPS)

unit. Furthermore, The sensing field is subdivided in a uniform way and the nodes are localised uniformly over the field. If the distance between two sensor nodes less than or equal to $2R_s$, Then they are called neighbours. The sensor disk model is used by both sensing and communication ranges.

3.2 Protocol idea

DiGALCO protocol is based on two steps. The sensing field is partitioned uniformly into smaller virtual subregions in the first step (Idrees et al., 2015). Second, DiGALCO protocol is then distributed and executed simultaneously in each sensor node in each subregion. Figure 1 shows proposed DiGALCO protocol.

Figure 1 Proposed DiGALCO protocol (see online version for colours)



As in the DiLCO protocol Idrees et al. (2015), DiGALCO protocol uses the same two packets during its implementation: INFO packet used for information exchange among sensor nodes and ActiveSleep packet used to inform sensor nodes to go Sleep or to remain Active during the monitoring time. Furthermore, every node in the network can be in one of the following five possible states: computation (sensor applies the optimisation process), active, sleep, listening and communication (transmitting or receiving packet). Figure 1 shows that the proposed DiGALCO protocol lifetime is divided into rounds, where each round is consists of three phases: discovery, cluster head election and sensor node activity scheduling-based GA optimisation, as well as monitoring. For each round, there will be only one schedule of sensor nodes responsible for monitoring the sensing field. This periodic scheduling is useful due to enhancement the robustness of the network against sensor node failures Idrees et al. (2015). The flowchart of DiGALCO protocol executed in each sensor node is presented in Figure 2.

As shown in Figure 2, if remaining power of $s_j \geq E_{th}$ (the minimum power required by the sensor to remain active for one round). Then, the sensor node will send and receive information packets to and from other nodes in the subregion. This packet contains sensor id, position and resedual power (RE_j). Once the first phase is completed, each sensor node in the subregion apply cluster head election procedure to elect the cluster head of the current subregion. Each s_j in the subregion applying equation (1) and for each sensor k , $k \in N_j$, where N_j is the overall nodes number in the subregion. The node will has a weight w_k to every node k in the subregion. The node k that has the maximum weight w_k will be elected as a head of cluster for this round. All the sensor nodes in the subregion will apply the same equation (1) and they will have the same winner as cluster head for the

subregion. After that, if the sensor node is cluster head, it will apply a GA to produce the best schedule for the current round (see Section 3.4).

$$w_k = \left(\alpha * \left(\frac{RE_k}{E_{\max}} \right) + \beta * \left(\sum_{i \in N_j} \left(1 - \frac{d(k, i)}{2 * R_s} \right) \right) \right) + z \quad (1)$$

$$z = \left(\frac{ID_k}{N_j} \right) * \delta, \alpha + \beta = 1, k \in N_j, \text{ and } \delta = 0.005$$

$$d(k, i) = \sqrt{(x_k - x_i)^2 + (y_k - y_i)^2} \quad (2)$$

As soon as the GA execution is terminated, the optimal sensor schedule is produced. The cluster head will send an ActiveSleep packet to each node in the subregion to inform it its new status. If the node does not represent a cluster head, it waits for ActiveSleep packet to change its status during monitoring time.

3.3 Coverage problem definition

In this section, the problem of coverage is mathematically modelled. The following notations are used throughout the section. In a subregion R, a set S of sensor nodes $S = \{s_1, s_2, \dots, s_i, \dots, s_{S_N}\}$, a set C of sensors centres $C = \{c_1, c_2, \dots, c_j, \dots, c_{S_N}\}$ and a set of alive sensors $A \subseteq S, A = \{s_1, s_2, \dots, s_k, \dots, s_{S_A}\}$. Find the best representative sensor node set X, $X = \{x_1, x_2, \dots, x_k, \dots, x_{S_A}\}$, that cover all the sensors centres in the subregion R with the minimum number of nodes in order to enhance the coverage lifetime of WSN.

$$\Psi_{k,j} = \begin{cases} 1 & \text{if node centre } j \text{ is covered by node } k, \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

$$X_k = \begin{cases} 1 & \text{if node } k \text{ is Active,} \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

$$\Upsilon_j = \sum_{k=1}^{S_A} (\Psi_{k,j} \cdot X_k) \quad (5)$$

Hence, the problem of lifetime coverage optimisation is modelled as follow.

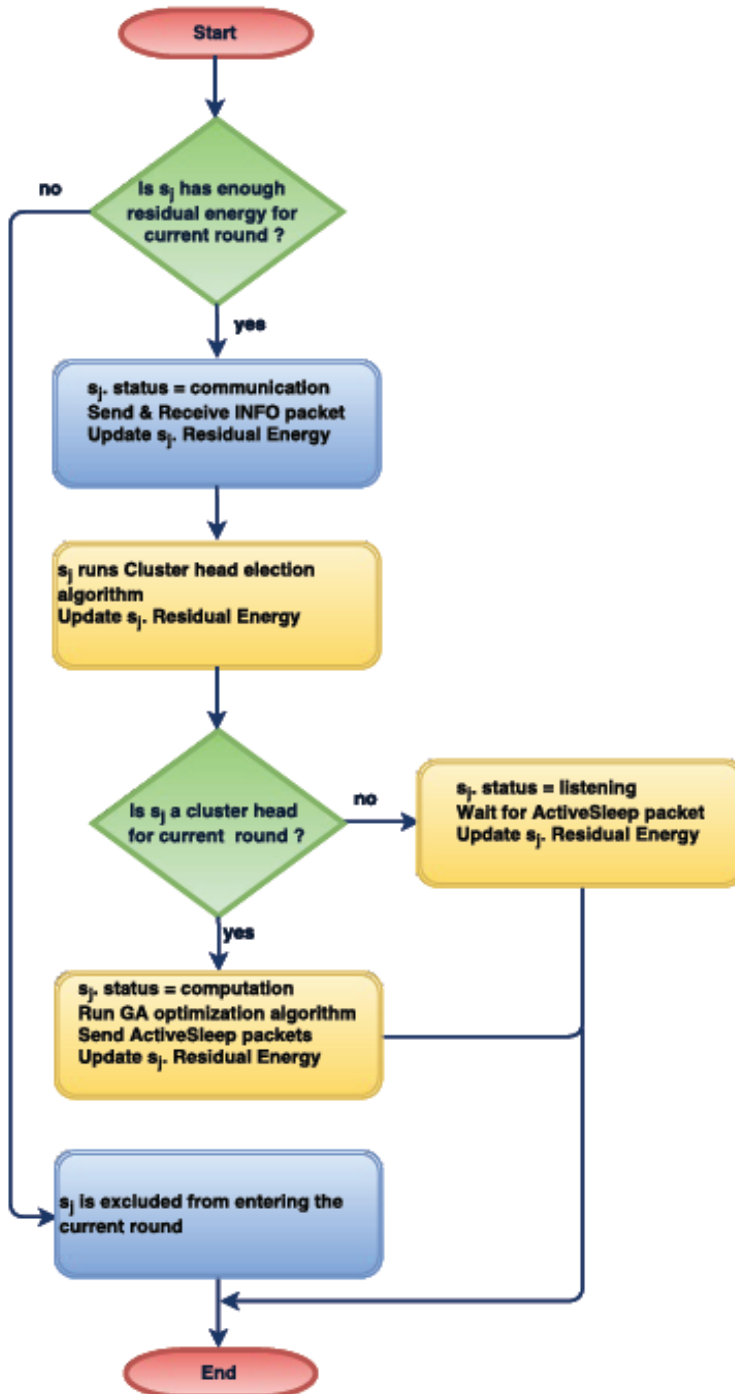
$$\min \sum_{j=1}^{S_N} \Upsilon_j \quad (6)$$

Subject to

$$\sum_{k=1}^{S_A} (\Psi_{k,j} \cdot X_k) = 1 \quad \forall j \in S_N \quad (7)$$

$$X_k \in \{0, 1\}, \forall k \in S_A \quad (8)$$

$$\Upsilon_j \in N, \forall j \in S_N \quad (9)$$

Figure 2 The flowchart of DIGALCO protocol (see online version for colours)

3.4 GA for lifetime coverage optimisation

Metaheuristics are a generic search strategies for exploring search spaces for solving the complex problems. These strategies have to dynamically balance between the exploitation and the exploration in both the stored experience of search and the space of search respectively. The solutions with the best quality can be determined in the space of search by using this balancing. They prevent consume a long time in the areas of the space of search that are searched or do not produce solutions with the best quality. Therefore, in order to solve the problem of optimisation and produce a good solution, the metaheuristic approach is the best candidate, especially with inadequate information or limited capacity of processing (Bianchi et al., 2009). GA is one of the population-based metaheuristic approaches that simulates the process of natural selection (Hassanien et al., 2015). GA starts with a population of random candidate solutions (called individuals or phenotypes). GA uses genetic operators inspired by natural evolution, such as selection, mutation, evaluation, crossover and replacement so as to improve the initial population of candidate solutions. This process repeated until a stopping criterion is satisfied. Since integer linear programming is NP-hard; (Sherali and Driscoll, 2000), many problem instances are complicated and so heuristic methods must be used instead. GA is a metaheuristic technique suitable to solve NP-hard problems and provide approximate solutions. In comparison with optimisation solvers, GA provides a near optimal solution with acceptable execution time, as well as it requires a less amount of memory especially for large size problems. Optimisation solvers provide optimal solution, but they require higher execution time and amount of memory for large problem.

In this section, we present a metaheuristic-based GA to solve proposed lifetime coverage optimisation problem. The proposed GA provides a near optimal schedule for sensing per round. The proposed GA is based on the mathematical model which is presented in Section 3.3. Algorithm 1 shows the proposed GA to solve the coverage lifetime optimisation problem. The new protocol which is based on GA in the decision phase is named as DiGALCO. The proposed GA can be explained in more details as follow.

The GA algorithm can be explained in more detail as follow:

- 1 *Representation*: Since the proposed GA's goal is to find the optimal (or near optimal) schedule of the sensor nodes which takes the responsibility of monitoring the subregion R in the next phase, a chromosome is defined as a schedule for S_A sensor nodes and each chromosome contains S_A genes. Therefore, the gene of such a chromosome is the schedule of a sensor. In other words, the genes corresponding to active nodes have the value of one, the others are zero. Figure 3 shows solution representation in the proposed GA.
- 2 *Initialise population*: The initial population is randomly generated and each chromosome in the GA population represents a possible sensors schedule solution to cover the entire subregion.
- 3 *Compute Υ_j and penalty function P_j* : After creating the initial population, each individual *ind* compute the parameters Υ_j^{ind} and P_j^{ind} [see equations (10) and (11)] so as to use them in the next step.

$$\Upsilon_j^{ind} = \begin{cases} 0 & \text{if } \sum_{k=1}^{S_A} (\Psi_{k,j}^{ind} \cdot X_k^{ind}) = 0, \\ \sum_{k=1}^{S_A} (\Psi_{k,j}^{ind} \cdot X_k^{ind}) & \text{otherwise.} \end{cases} \quad (10)$$

$$P_j^{ind} = \begin{cases} S_N & \text{if } \sum_{k=1}^{S_A} (\Psi_{k,j}^{ind} \cdot X_k^{ind}) = 0, \\ 0 & \text{otherwise.} \end{cases} \quad (11)$$

- 4 *Evaluate population:* After creating the initial population, each individual is evaluated and assigned a fitness value according to the fitness function that illustrated in equation (12). In the proposed GA, the optimal (or near optimal) candidate solution, is the one with the minimum value for the fitness function. The lower the fitness values been assigned to an individual, the better opportunity it get survived. In the proposed work, the function rewards the decrease in the sensor nodes which cover the same sensor centre and penalises the decrease to zero in the sensor nodes which cover the sensor centre.

$$F^{ind} \leftarrow \sum_{j=1}^{S_N} (\Upsilon_j^{ind} + P_j^{ind}) \quad (12)$$

- 5 *Selection:* In order to generate a new generation, a portion of the existing population is elected based on a fitness function that ranks the fitness of each candidate solution and preferentially select the best solutions. A best individual selection (BIS) is used by the proposed GA. By using BIS, a best two individuals are selected from the current population based on fitness values of individuals. The selected best two individuals sent to the mating pool.
- 6 *Crossover:* Crossover is a genetic operator used to take more than one parent solutions and produce a child solution from them. If crossover probability P_c is 100%, then the crossover operation takes place between two individuals. If it is 0%, the two selected individuals in the mating pool will be the new chromosomes without crossover. In the proposed GA, a two-point crossover is used. The two-point crossover is explained in Figure 4.
- 7 *Mutation:* Mutation is a divergence operation which introduces random modifications. The purpose of the mutation is to maintain diversity within the population and prevent premature convergence. Mutation is used to add new genetic information (divergence) in order to achieve a global search over the solution search space and avoid to fall in local optima. The mutation operator in the proposed GA works as follow: If mutation probability P_m is 100%, then the mutation operation takes place on the new individual. Two different indices are selected randomly within the schedule solution. After that, the contents of the positions are exchanged. If mutation probability P_m is 0%, then the mutation operation does not take place.
- 8 Before evaluating each new individual, update Υ^{ind1} , P^{ind1} , Υ^{ind2} , and P^{ind2} of $Child^{ind1}$ and $Child^{ind2}$ respectively using equations (10) and 11.

- 9 *Evaluate new individuals*: After updating the parameters Υ_j and P_j , each new individual is evaluated using equation (12).
- 10 *Replacement*: After evaluation of new children, Triple Tournament Replacement (TTR) will be applied for each new individual. In TTR strategy, three individuals are selected randomly from the population. Find the worst from them and then check its fitness with the new individual fitness. If the fitness of the new individual is better than the fitness of the worst individual, replace the new individual with the worst one. Otherwise, the replacement is not done.
- 11 *Stopping criteria*: The proposed GA ends its execution as soon as one of two stopping criteria is reached, which make use of two parameters, called MaxIter and MaxMinFit respectively. The MaxIter parameter refers to a maximum number of iterations that performed by the GA. The MaxMinF it is met when the maximum fitness value in the population is equal to the minimum fitness value. The best solution will be chosen as a schedule of sensor nodes for the sensing phase in the current round.

Algorithm 1 $GA(s_j)$

Input: S_{pop} , P_c , P_m and S_A

Output: $X = \{x_1, x_2, \dots, x_k, \dots, x_{S_A}\}$, the schedule of sensor nodes

foreach *Individual* $ind \in S_{pop}$ **do**

 Generate Randomly Chromosome X ;

 Compute Υ^{ind} and P^{ind} ;

 Evaluate Individual X^{ind} ;

end

while *Stopping criteria is not satisfied* **do**

 Selection ($ind1, ind2$);

 Crossover

 ($P_c, X^{ind1}, X^{ind2}, Child^{ind1}, Child^{ind2}$);

 Mutation ($P_m, Child^{ind1}, Child^{ind2}$);

 Update Υ^{ind1} and P^{ind1} of $Child^{ind1}$;

 Update Υ^{ind2} and P^{ind2} of $Child^{ind2}$;

 Evaluate New Individuals $Child^{ind1}$ and $Child^{ind2}$;

 Replacement ($Child^{ind1}, Child^{ind2}$);

end

$\{(X_1, \dots, X_k, \dots, x_{S_A})\} = \text{Select Best Solution}$

(S_{pop});

return X ;

Figure 3 Solution representation by the proposed GA, (a) sensors status inside the subregion (b) candidate solution represented by chromosome for the sensors in (a) (see online version for colours)

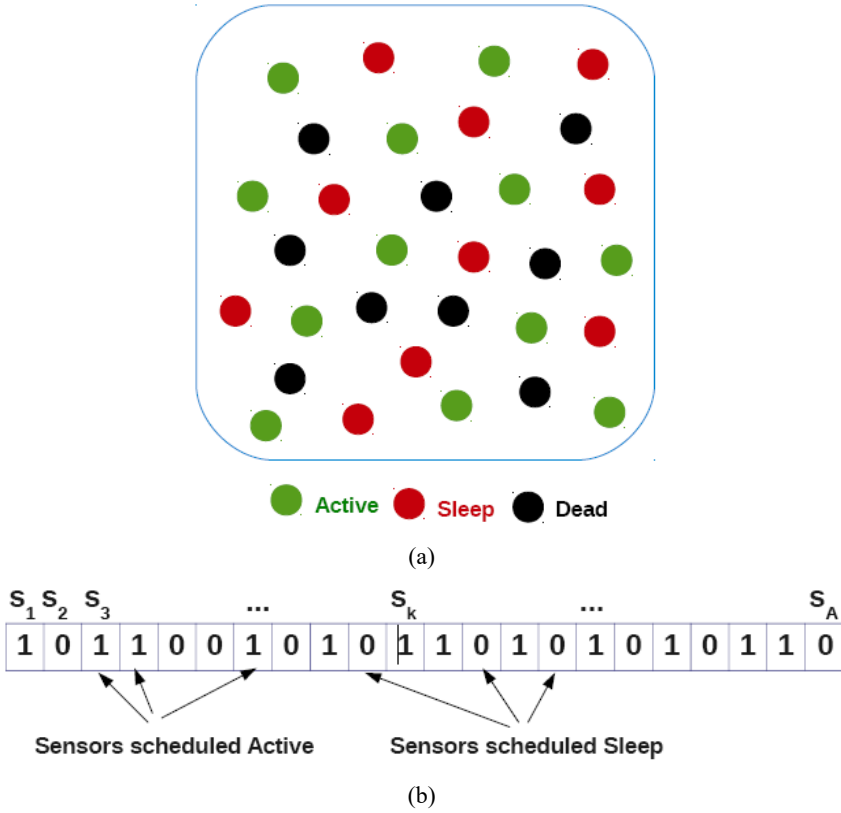
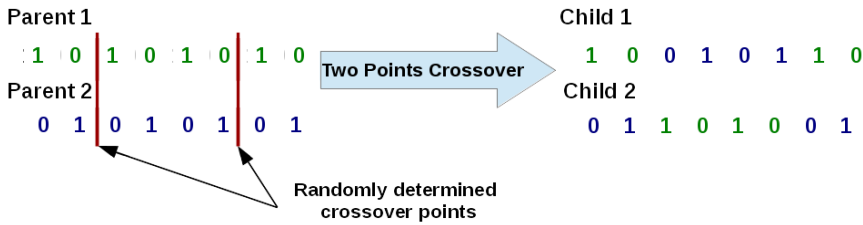


Figure 4 Two-point crossover (see online version for colours)



4 Performance evaluation and analysis

The performance of DiGALCO protocol is evaluated by achieving several experiments using OMNeT++ network simulator (Varga, 2003). Table 1 illustrates the parameters used in this simulation.

Fifty executions are achieved using various WSN topologies. The introduced results are an average of these 50 executions. Five network sizes (100 to 300 nodes) are used by

the simulations. The nodes are deployed in a controlled way over a sensing field of $(50 \times 25) m^2$ to ensure initial full coverage with current sensing range. The sensing field of WSN is assumed to be partitioned into 16 regular subregions and the proposed protocol used the same energy model that explained in Idrees et al. (2015).

Table 1 Parameters used during the simulations

<i>Parameter</i>	<i>Value</i>
Sensing field	$(50 \times 25) m^2$
Nodes number	100, 150, 200, 250 and 300 nodes
Initial energy	500–700 Joules
Sensing period	60 minutes
E_{thr}	36 Joules
R_s	5 m
R_c	10 m
P_c	0.95
P_m	0.5
MaxIter	1,000
S_{pop}	50

Each sensor node initialised its power randomly within the range [500–700]. In the experimentations, the same performance metrics that used by Idrees et al. (2015) are used to evaluate the efficiency of DiGALCO protocol. These metrics are coverage ratio (CR), active sensors ratio (ASR), network lifetime and energy consumption. Besides DiGALCO, four other methods are used for the purpose of comparison. These methods are DESK (Vu et al., 2006), GAF (Xu et al., 2001), DiLCO protocol (Idrees et al., 2015) and PeCO protocol (Idrees et al., 2016). The same framework is used by DiGALCO, PeCO and DiLCO protocols. These protocols are used different mathematical models and optimisation approaches so as to provide the best schedule of sensor nodes for monitoring in the sensing phase. DiLCO protocol attempts to ensure the full area coverage by covering a set of points in the sensing field. PeCO protocol tries to provide a certain coverage level for every sensor perimeter. DiLCO and PeCO used an integer program that implemented by the GLPK optimisation solver. DiGALCO protocol attempts to fulfil the coverage of centre points of sensor nodes. DiGALCO protocol uses GA to optimise both coverage and network lifetime.

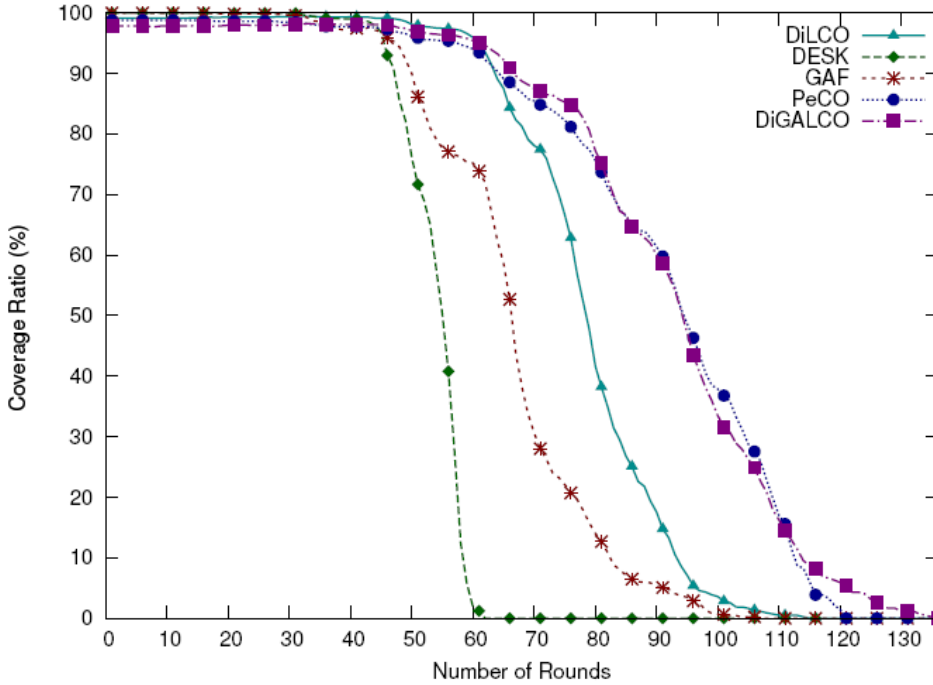
4.1 Coverage ratio

The average ratio of coverage of the five protocols for 200 nodes is shown in Figure 5. For the first rounds, DESK, GAF and DiLCO produce a lightly better ratio of coverage (with respectively 99.99%, 99.91% and 99.02%), in comparison with the (98.76% and 97.88%) provided by PeCO and DiGALCO respectively. The reason behind this is that DiGALCO, PeCO and DiLCO turn off a larger number of redundant nodes compared with GAF and DESK protocols. After the round 33th, DiGALCO protocol gives better coverage performance compared with other approaches and maintains a level of coverage bigger than 80% for a larger number of rounds. This increased performance of coverage caused by a large amount of saved energy by DiGALCO in the first rounds.

4.2 Active sensors ratio

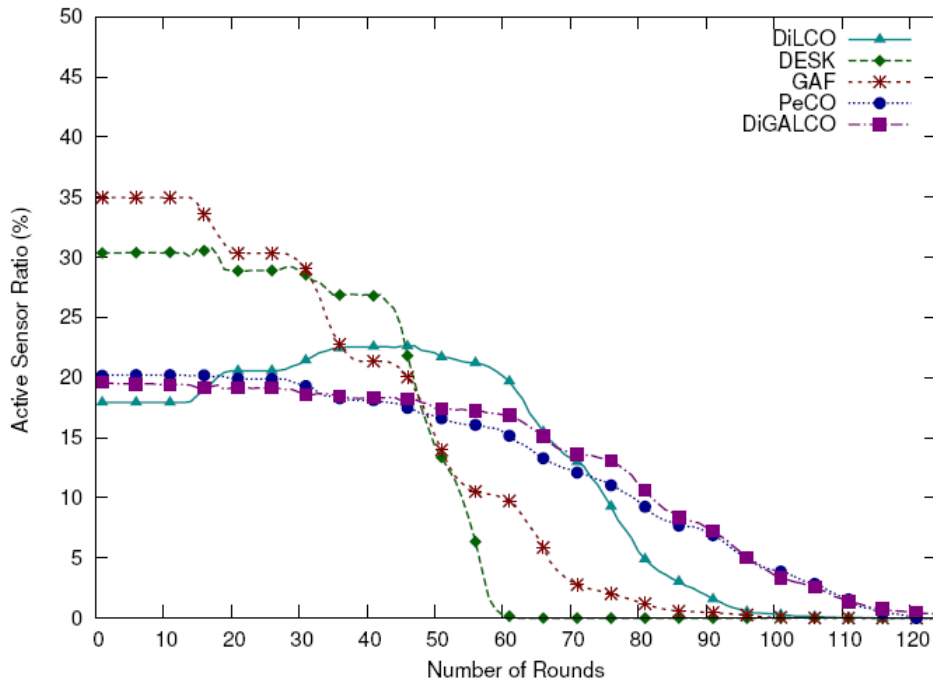
In each round, it is necessary to decrease the active nodes number in order to conserve a large amount of power and consequently to maximise the WSN lifetime. Figure 6 exhibits the ratio of average active nodes for 200 deployed nodes. In the first fourteen rounds, DESK and GAF activate 30.36 % and 34.96 % nodes whilst DiLCO, PeCO and DiGALCO activate 17.92 %, 20.16 % and 19.64 % active sensor nodes respectively. DiGALCO protocol activates a higher nodes number when the rounds number increases compared with PeCO, DESK and GAF in order to provide a higher level of coverage as displayed in Figure 5.

Figure 5 CR for 200 deployed nodes (see online version for colours)



4.3 Energy consumption

This section presents the impact of energy consumed by the network during the different status of sensor nodes (such as communication, computation, listening, active and sleep status) for various WSN sizes and the comparison with other approaches is investigated. Figures 7(a) and 7(b) exhibit the consumed energy for various WSN sizes and for $Lifetime_{95}$ and $Lifetime_{50}$. The comparison illustrates the superiority of DiGALCO in the power saving. The two figures show the reduced consumption of energy by DiGALCO compared with other approaches. When the network size is greater than 200 nodes, the consumed energy is the lowest with PeCO for $Lifetime_{50}$.

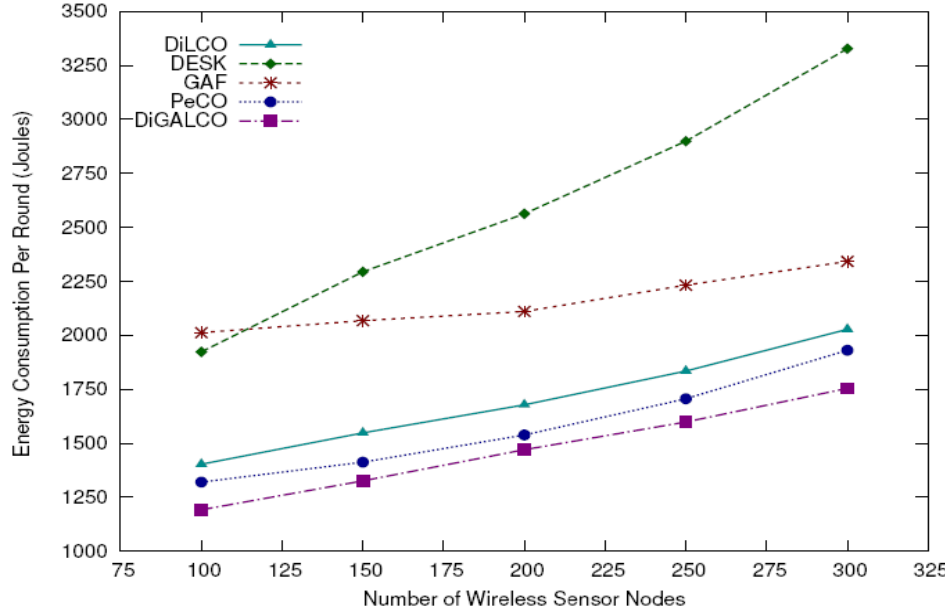
Figure 6 ASR for 200 deployed nodes (see online version for colours)

4.4 Network lifetime

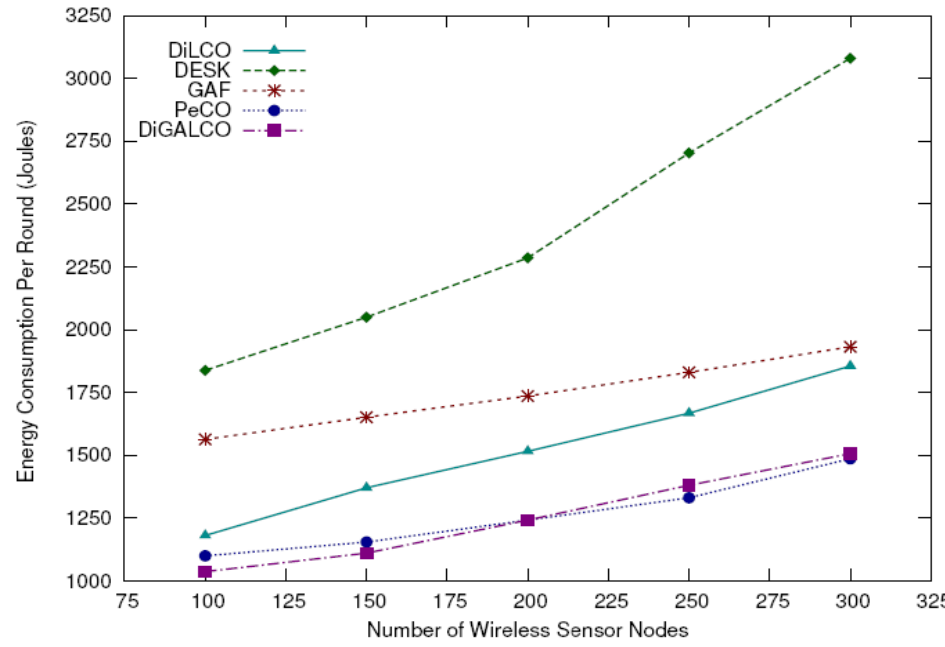
The comparison results show that DiGALCO and DiLCO provide a better lifetime improvement for $Lifetime^{95}$, whilst DiGALCO and PeCO give the best results for $Lifetime^{50}$. In Figures 8(a) and 8(b), $Lifetime^{95}$ and $Lifetime^{50}$ are exhibited for various WSN densities. These figures show that DiGALCO, DiLCO and PeCO enhance the network lifetime when the WSN size increases. In the network lifetime for $Lifetime^{95}$ [see Figure 8(a)], DiGALCO protocol can enhance the network lifetime up to 4.7%, 32.6%, 24.4% and 14% compared to DiLCO, DESK, GAF and PeCO protocols respectively. In the network lifetime for $Lifetime^{50}$ [see Figure 8(b)], DiGALCO protocol can enhance the network lifetime up to 17.2%, 46%, 25.4% compared to DiLCO, DESK and GAF protocols respectively, Whilst, PeCO protocol enhance the network lifetime up to 2.4% compared to DiGALCO protocol.

Figure 9 shows the comparison results among three protocol (DiLCO, PeCO and DiGALCO) from the coverage lifetime point of view and for various levels of coverage. The network able to provide a coverage level higher than 80%, 85%, 90% and 95% by Protocol/80, Protocol/85, Protocol/90 and Protocol/95 respectively, where the word Protocol refers to DiGALCO, PeCO, or DiLCO. In fact, some applications do not need a full coverage in all the time, for instance, forest fire application requires only 80% coverage level in the rainy seasons whilst it requires a full coverage in the summer seasons Li et al. (2011). DiGALCO always outperforms PeCO and DiLCO for all levels of coverage. Furthermore, the enhancement increases when the network size increases.

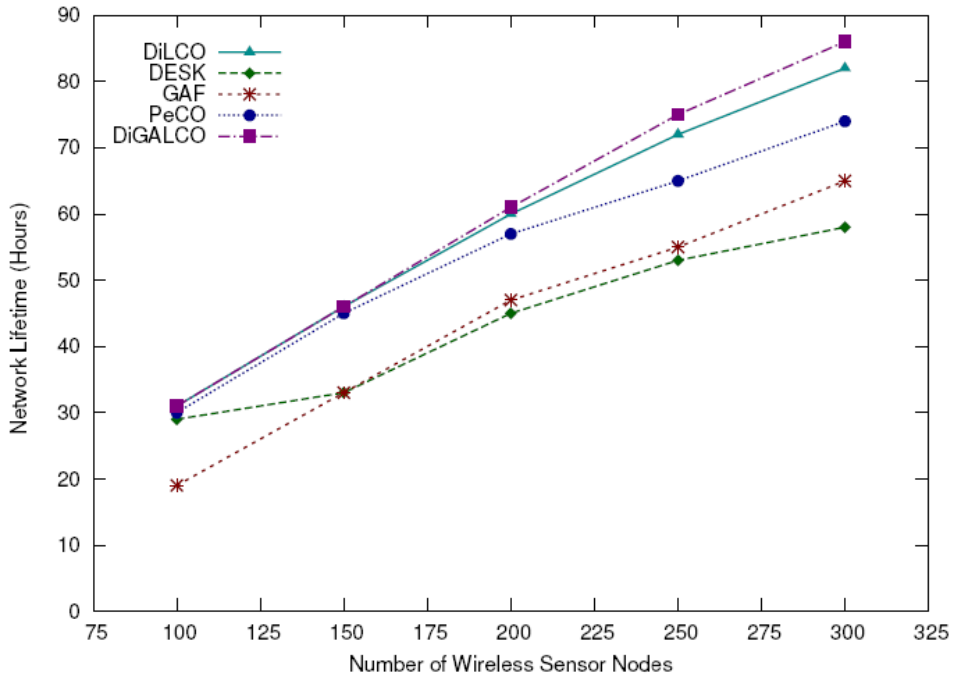
Figure 7 Energy consumption per period for (a) *Lifetime₉₅* and (b) *Lifetime₅₀* (see online version for colours)



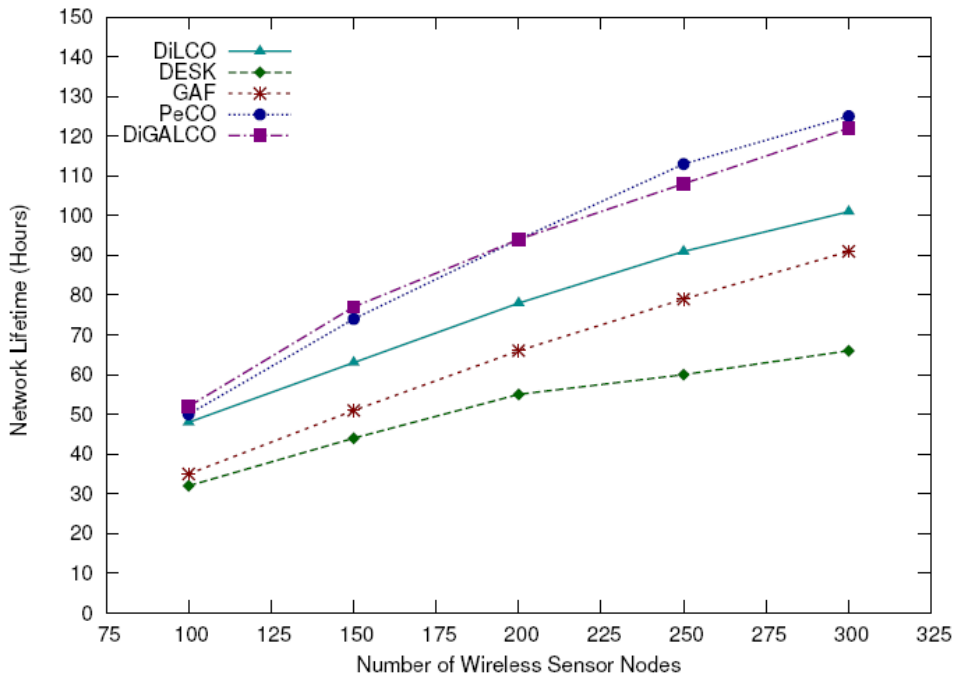
(a)



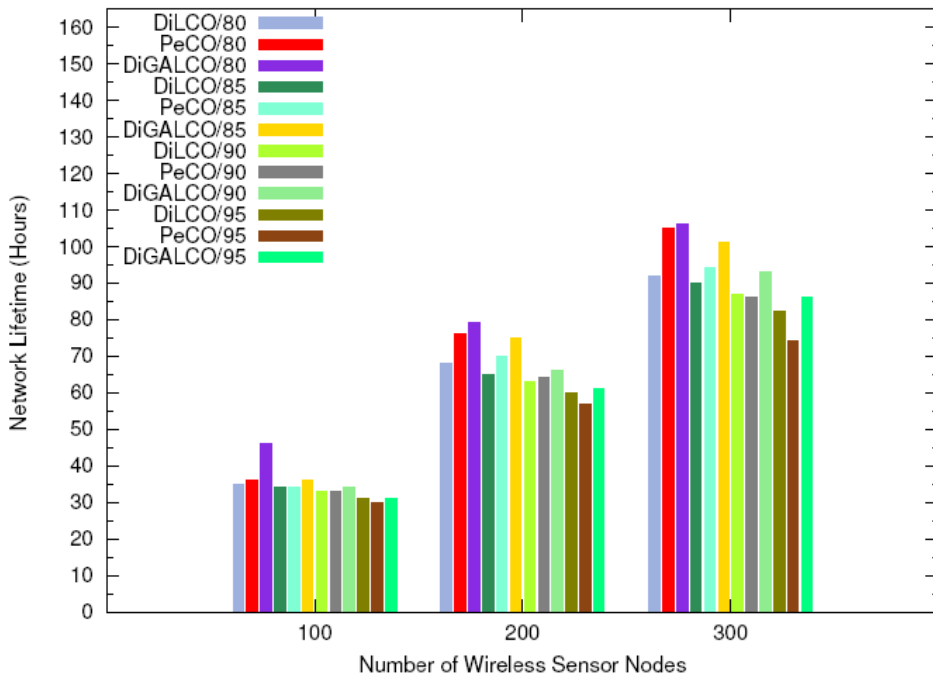
(b)

Figure 8 Network lifetime for (a) *Lifetime₉₅* and (b) *Lifetime₅₀* (see online version for colours)

(a)



(b)

Figure 9 Network lifetime for different CRs (see online version for colours)

4.5 Discussions

This section discuss the feasibility of our protocol in a real limited resources wireless sensor better than existing distributed techniques. The average execution time is required to solve our proposed coverage optimisation problem is reported for several methods and different network sizes. We compute the original execution time on a DELL laptop with Intel Core i3 2370 M (2.4 GHz) processor (2 cores) and the MIPS (Million Instructions Per Second) rate equal to 35330. In order to be compatible with the use of a sensor node with Atmels AVR ATmega103L microcontroller (6 MHz) and a MIPS rate equal to 6 to run the optimisation resolution, this time is multiplied by $\left(\frac{35330}{2} \times \frac{1}{6}\right)$ and reported on

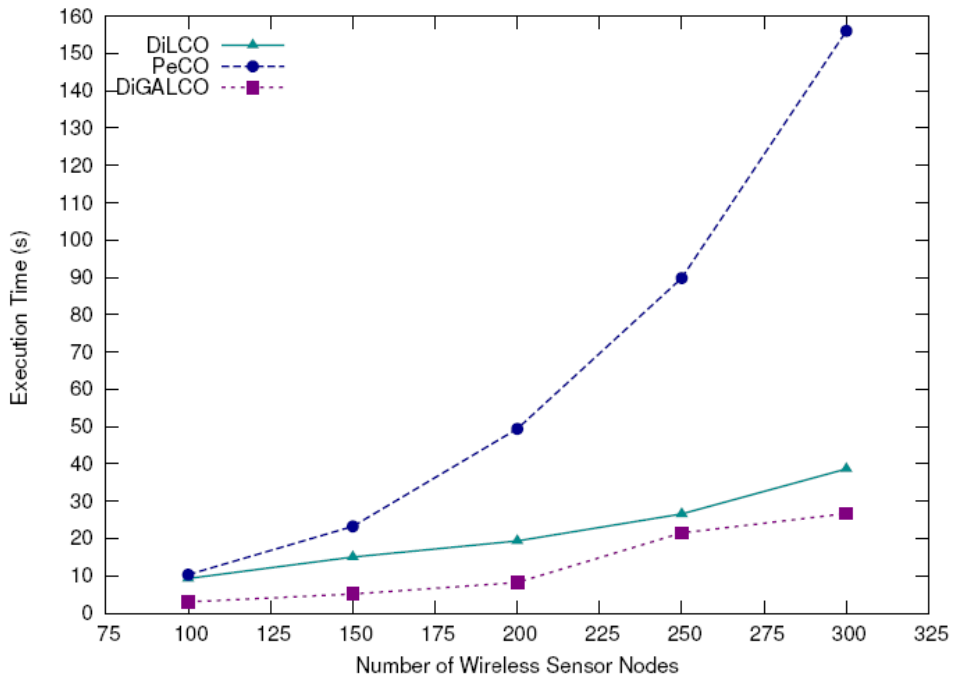
Figure 10.

Figure 10 shows that DiGALCO protocol has lower execution times in comparison with other methods due to utilising a metaheuristic method (GA) to find good approximate solutions for the coverage optimisation problem instead of using the optimisation solver.

In practice, executing GLPK (GNU linear Programming Kit available in the public domain) on a sensor node is obviously intractable because it consumes a huge memory inside the sensor node. Therefore, a lightweight heuristic can be a good candidate to solve the optimisation problem. From the space complexity point of view, DiGALCO protocol takes $O(S_A S_{pop})$ as space complexity. If we consider a WSN of 300 nodes, the number of sensor nodes in each subregion is 18.5. Therefore, each cluster head in the subregion will execute the GA with a maximum number of sensor nodes equal to 19 nodes. In our

simulation, we represent the chromosome as a vector of integer values and we consider S_{pop} to 50. The memory needed by our protocol is composed of two parts: the constant space plus the variable space. The variable part of space is nearly equal to 1.9 KB ($19 * 2 \text{ bytes} * 50 = 1,900 \text{ bytes}$). Since each sensor node (Medusa II) which uses Atmels AVR ATmega103L microcontroller has 128 KB of memory, therefore, this can ensure that our proposed GA can be executed by real sensor node.

Figure 10 Execution time (in seconds) (see online version for colours)



DiGALCO protocol optimise both the coverage and lifetime in each subregion of the WSN. It is a hybrid of centralised and distributed methods. It is a distributed on the subregions obtained by dividing the area of interest into subregions and it achieves the optimisation at each cluster head in the subregions to provide the best schedule of sensor nodes in a centralised way to preserve the coverage and to enhance the WSN lifetime. To be acceptable for large-scale networks, the sensing field in the DiGALCO protocol is partitioned into many smaller subregions. One cluster head will be selected to run the optimisation algorithm (GA) to provide the active sensors for the current round. Therefore, DiGALCO protocol is scalable and a globally distributed approach, whilst it is centralised in each subregion. In the full distributed approaches, the decision is based only on the local information from the neighbouring nodes and there is no optimisation. This is why our DiGALCO protocol is better than existing distributed techniques because of applying a distributed optimisation on the subregions. The comparison results which are explained in the Sections 4.1–4.4 ensure the superiority of our proposed DiGALCO protocol in comparison with other existing techniques.

5 Conclusions and future works

The optimisation of lifetime coverage problem in sensor networks has been investigated in this article. A protocol named DiGALCO for maintaining the coverage and enhance the lifetime is proposed. DiGALCO is responsible for scheduling the sensor nodes activities periodically. DiGALCO is distributed inside the nodes in the subregions that obtained after dividing the sensing field in the first phase. It works into rounds and is based on the GA to choose the schedule of sensor nodes operating in active status for each round. In a subregion, DiGALCO is responsible for selecting the cluster head sensor node which will then apply a GA for activity scheduling of sensor nodes. The main challenges include how to elect the cluster head efficiently in the subregion as well as selecting the best schedule of sensor nodes to maintain a good CR. Several simulations have been performed to assess the performance of DiGALCO. The results illustrate that DiGALCO protocol is better than other methods in terms of lifetime, coverage level, active nodes ratio and consumed energy. In future, we plan to extend our work so that the optimisation algorithm provides multiple sensing rounds schedule. In addition, the optimisation model can be improved to consider into account the heterogeneity of the nodes in the network.

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